EXPERIMENT NO: - 01

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AIM: - To study various wireless technologies, their working principles, frequency ranges, and applications in modern communication.

WIRELESS TECHNOLOGIES

1. Bluetooth

Bluetooth technology allows devices to communicate with each other without cables or wires. Bluetooth relies on short-range radio frequency, and any device that incorporates the technology can communicate as long as it is within the required distance. The technology is often used to allow two different types of devices to communicate with each other. It is electronics "standard," which means that manufacturers that want to include this feature have to incorporate specific requirements into their electronic devices. These specifications ensure that the devices can recognize and interact with other devices that use Bluetooth technology.

Many personal electronic devices (PEDs) use Bluetooth technology. For example, you may be able to operate your computer with a wireless keyboard or use a wireless headset to talk on your mobile phone.

- **Range:** 1 to 100 meters, depending on the class (Class 1 offers the longest range).
- **Frequency:** Operates in the 2.4 GHz ISM (Industrial, Scientific, and Medical) band.
- **Applications:** Commonly used for connecting personal devices like wireless headphones, keyboards, mice, speakers, smartwatches, and file sharing between devices.
- **Standards:** Versions range from Bluetooth 1.0 to the latest Bluetooth 5.2 (and beyond), with newer versions offering improved speed, range, and connectivity features like mesh networking.

2. Wifi

A Wi-Fi network is simply an internet connection that’s shared with multiple devices in a home or business via a wireless router. The router is connected directly to your internet modem and acts as a hub to broadcast the internet signal to all your Wi-Fi enabled devices. This gives you

flexibility to stay connected to the internet as long as you're within your network coverage area.
Wi-Fi uses radio waves to transmit data from your wireless router to your Wi-Fi enabled devices like your TV, smartphone, tablet and computer.
Because they communicate with each other over airwaves, your devices and personal information can become vulnerable to hackers, cyber-attacks and other threats. This is especially true when you connect to a public Wi-Fi network at places like a coffee shop or airport. When possible, it's best to connect to a wireless network that is password-protected or a personal hotspot.

- **Range**: Typically around 30 meters indoors, but can extend further with Wi-Fi extenders or mesh systems.
- **Frequency**: Operates in 2.4 GHz, 5 GHz, and increasingly in 6 GHz bands (Wi-Fi 6/6E).
- **Applications**: Internet access, streaming, file sharing, home automation, and more. Used in homes, offices, cafes, airports, and public spaces.
- **Standards**: The latest standard is Wi-Fi 6 (802.11ax), which improves data rates, efficiency, and supports more devices simultaneously. Wi-Fi 5 (802.11ac) is also widely used.

3. **ZigBee**

Zigbee is a wireless technology developed as an open global market connectivity standard to address the unique needs of low-cost, low-power wireless IoT data networks. It is designed to support wireless communications, monitoring, and control of battery-operated devices and sensor networks. Zigbee operates on the IEEE 802.15.4 specification and is widely used in home automation, medical data collection, and industrial control systems.
The specification is a packet-based protocol intended for low-cost, battery-operated devices and products. The Zigbee IoT protocol allows devices to communicate data seamlessly in a variety of network topologies using extremely limited power. Thanks to Zigbee, devices from different manufacturers can communicate.

- **Range**: 10 to 100 meters, depending on the environment and network topology (Zigbee can form mesh networks).
- **Frequency**: Operates in the 2.4 GHz, 868 MHz, and 915 MHz frequency bands.
- **Applications**: Home automation (e.g., smart bulbs, smart locks), industrial automation, health monitoring, and agricultural sensors.
- **Standards**: Zigbee 3.0 is the latest version, supporting a wide range of device types and applications.

4. LoRa

LoRa stands for Long Range Radio and is mainly targeted for the Internet of Things (IoT) and M2M networks. It is a sort of new wireless modulation approach designed precisely for long-range connectivity and low-power communications. This technology will allow multi-tenant or public networks to connect a number of applications running on the same network.

The LoRa protocol includes a number of different layers including application and device-level for secure communications, encryption at the network. Each individual LoRa gateway has the capability to handle up to millions of nodes. The signals can extend a significant distance, which means that there is less structure required, making constructing a network faster and much cheaper to implement.

- **Range**: Up to 15 km in rural areas, and 2-5 km in urban environments.
- **Frequency**: Operates in sub-GHz frequencies (868 MHz, 915 MHz, or 433 MHz depending on the region).
- **Applications**: IoT networks like agriculture (soil moisture monitoring), smart cities (parking and waste management), asset tracking, and environmental monitoring.
- **Standards**: LoRaWAN (Long Range Wide Area Network) is the network layer protocol for LoRa, enabling large-scale IoT networks.

5. 5G

5G is the fifth-generation mobile network technology, designed to provide faster speeds, lower latency, and more reliable connections than previous generations.5G technology has a theoretical peak speed of 20 Gbps, while the peak speed of 4G is only 1 Gbps. 5G also promises lower latency, which can improve the performance of business applications as well as other digital experiences (such as online gaming, videoconferencing, and self-driving cars).
While earlier generations of cellular technology (such as 4G LTE) focused on ensuring connectivity, 5G takes connectivity to the next level by delivering connected experiences from the cloud to clients. 5G networks are virtualized and software-driven, and they exploit cloud technologies.
The 5G network will also simplify mobility, with seamless open roaming capabilities between cellular and Wi-Fi access. Mobile users can stay connected as they move between outdoor wireless connections and wireless networks inside buildings without user intervention or the need for users to reauthenticate.

- **Range**: Coverage can vary depending on the frequency used. Low-band 5G offers broader coverage, while high-band (mmWave) 5G provides faster speeds but has shorter range (up to a few hundred meters).
- **Frequency**: Uses a wide range of frequency bands, from sub-1 GHz to mmWave (24 GHz and higher).

- **Applications**: Ultra-fast mobile internet, augmented reality (AR), virtual reality (VR), autonomous vehicles, IoT devices, and smart cities.
- **Standards**: 5G New Radio (NR) is the standard that defines the air interface and technology for 5G deployment.

6. NFC

Near Field Communication (NFC) is a short-range wireless technology that enables communication between two electronic devices over a distance of 4 centimeters (1.6 inches) or less.
NFC relies on inductive coupling between two electromagnetic coils present on NFC-enabled devices (such as smartphones). Communication occurs at a frequency of 13.56 MHz within the globally available unlicensed radio frequency ISM band. Data rates range from 106 to 848 kbit/s. NFC can be used for various purposes, including contactless transactions, data exchange, and simplified setup of more complex communications (e.g., Wi-Fi). When one of the connected devices has internet connectivity, data exchange with online services is also possible. The NFC standard currently has three distinct modes of operation to determine what sort of information will be exchanged between devices.

- **Range**: Extremely short-range communication (4 cm or less).
- **Frequency**: Operates at 13.56 MHz.
- **Applications**: Contactless payments (e.g., Apple Pay, Google Wallet), ticketing, data transfer between devices, and pairing of devices like Bluetooth speakers.
- **Standards**: NFC is defined by ISO/IEC 14443 and ISO/IEC 18092 standards.

7. Infrared (IR)

Infrared radiation (IR), sometimes referred to simply as infrared, is a region of the electromagnetic radiation spectrum where wavelengths range from about 700 nanometers (nm) to 1 millimeter (mm).
Infrared waves are longer than visible light waves but shorter than radio waves. Correspondingly, the frequencies of IR are higher than microwave frequencies but lower than visible light frequencies, ranging from about 300 gigahertz to 400 terahertz (THz).
Infrared light is invisible to the human eye, but heat sensors can detect longer infrared waves. Infrared shares some characteristics with visible light, however. Like visible light, infrared light can be focused, reflected and polarized.

- **Range**: Typically up to 5 meters, though it can be extended with line-of-sight and high-power devices.

- **Frequency**: IR devices operate in the 30-300 THz range.
- **Applications**: Remote controls, short-range data transfer between devices, and IR-based sensors.
- **Standards**: IR protocols like IrDA (Infrared Data Association) allow for data exchange between devices, though this has largely been superseded by Bluetooth.

8. Satellite Communication

Satellite communication allows for long-range communication via satellites orbiting Earth, making it ideal for remote or underserved areas.
Satellite communication is transporting information from one place to another using a communication satellite in orbit around the Earth. Watching the English Premier League every weekend with your friends would have been impossible without this. A communication satellite is an artificial satellite that transmits the signal via a transponder by creating a channel between the transmitter and the receiver at different Earth locations.

- **Range**: Global, depending on the satellite network's coverage and location.
- **Frequency**: Uses various frequency bands, including L-band, C-band, Ku-band, and Ka-band.
- **Applications**: Global communications, satellite internet, broadcasting, GPS, weather monitoring, and military applications.
- **Standards**: Different satellite communication systems, such as GEO (Geostationary Earth Orbit), LEO (Low Earth Orbit), and MEO (Medium Earth Orbit), each offering different trade-offs in latency and coverage.

9. Cellular (2G, 3G, 4G, 5G)

Cellular networks provide mobile communication (voice, text, and data) to users via cell towers in large geographic areas. Each generation improves on speed, latency, and capacity.

- **Range**: Varies from a few kilometers in urban areas (for 4G and 5G) to tens of kilometers in rural or low-density areas.
- **Frequency**: Operates in multiple frequency bands, ranging from 700 MHz to 100 GHz (in the case of 5G).
- **Applications**: Voice calls, mobile internet, video streaming, IoT applications, and emergency communication.
- **Standards**: 2G (GSM), 3G (UMTS, CDMA), 4G (LTE), and 5G (New Radio)

PARAMETERS

1. **Range of Communication**:

   This refers to the maximum distance over which a device or technology can communicate effectively. For wireless technologies, the range depends on factors like signal strength, environmental conditions, and interference. For example, Wi-Fi might have a range of a few hundred feet indoors, while cellular networks can cover miles.

2. **Cost of the Devices for that Technology**:

   The cost refers to the price of the devices required for communication using a specific technology. This could include the cost of transmitters, receivers, routers, or specialized devices (e.g., 5G base stations). Higher technology sophistication usually correlates with higher device costs.

3. **Propagation Delay**:

   Propagation delay is the time it takes for a signal to travel from the sender to the receiver. This delay is influenced by the distance between the devices and the medium through which the signal travels. For instance, fiber-optic networks have lower propagation delays compared to satellite communication due to the distance signals must travel.

4. **Power Consumption**:

   Power consumption refers to the amount of electrical energy a device or technology uses while in operation. Efficient power use is crucial for mobile devices and large-scale networks, especially for IoT devices, which often rely on battery power.

5. **Throughput**:

   Throughput is the rate at which data is successfully transmitted from one point to another over a network or communication system. It is usually measured in bits per second (bps) or multiples like Mbps or Gbps. Higher throughput is essential for applications that require large amounts of data to be transferred quickly, such as video streaming or cloud services.

6. **Fan Out**:

   Fan out refers to the number of devices or users a single device or communication node can serve or connect to. In networking, a higher fan-out indicates that a node or device can handle a larger number of simultaneous connections.

➢ COMPARISION OF TECHNOLOGIES

Technology	Range of Communication	Cost of the Devices	Propagation Delay	Power Consumption	Throughput	Fan Out
Bluetooth	Up to 100 meters	Low	Low	Low	Up to 3 Mbps	Moderate
Wi-Fi	100-300 meters (indoor)	Moderate	Low	Moderate	50 Mbps - 1 Gbps	Moderate to High
Zigbee	10-100 meters	Very Low	Low	Very Low	20-250 kbps	Moderate
LoRa	Up to 15 km (rural)	Low	Moderate	Very Low	0.3-27 kbps	Low
5G	100 meters - 1 km (indoor)	High	Low	High	1 Gbps - 10 Gbps	Very High
NFC	Up to 10 cm	Very Low	Very Low	Very Low	424 kbps	Low
Infrared	1-10 meters	Low	Low	Low	4-16 kbps	Low
Satellite Communication	Up to thousands of km	High	High	High	1 Mbps - 10 Gbps	Very Low
Cellular	1-100 km	Moderate to High	Moderate	Moderate	10 Mbps - 1 Gbps	High

Conclusion: -

Wireless technologies play a crucial role in modern communication by enabling seamless data transfer without physical connections. Each technology serves a unique purpose, from short-range communication (NFC, Bluetooth, ZigBee) to long-range applications (LoRa, Satellite Communication, 5G). Understanding these technologies helps in selecting the right one based on factors like range, frequency, power consumption, and application requirements.